AI-Powered Real Estate Intelligence, Valuation & Investment Recommendation Platform

Data Cleaning Report

Type	Regression + Recommendation System — Real Estate Intelligence Platform
Dataset	Indian Real Estate Housing Dataset (Kaggle) + Census Population Density Data
Dataset URL	kaggle.com/datasets/ankushpanday1/india-house-price-prediction
Records	2,50,000 Property Records
Features	34 Features (23 Original + 11 Engineered)
Best Model	XGBoost Regressor
Test Accuracy	R^2 Score = 0.9769
MAE | RMSE	13.68 Lakhs | 18.03 Lakhs
Deployment	Streamlit web application
Notebooks	01_data_inspection_&_cleaning

This Data Cleaning Report documents the preprocessing and quality assurance steps performed on the Indian Real Estate dataset used for the project. The cleaning process included handling missing values, removing duplicate records, validating data consistency, and addressing outliers to improve dataset reliability. The resulting clean dataset provides a high-quality foundation for exploratory analysis, feature engineering, machine learning model development, and deployment of the AI-powered real estate intelligence platform.

1. Introduction

The objective of the data cleaning phase is to ensure the dataset is accurate, consistent, and ready for exploratory data analysis and machine learning. Raw data often contains missing values, duplicates, outliers, and logical errors that can negatively impact model performance. This report details the comprehensive data quality assessment and the preprocessing steps applied to the real estate dataset.

The primary dataset was sourced from Kaggle ("India House Price Prediction"). The raw dataset contains records representing real estate properties across various cities in India, capturing details like property size, price, location, and amenities.

2. Initial Data Assessment

An initial inspection of the raw dataset revealed the following dimensions:

- **Dataset Shape:** 250,000 rows × 23 columns

Column Names and Data Types:

Column Name	Data Type
ID	int64
State	object (String)
City	object (String)
Locality	object (String)
Property_Type	object (String)
BHK	int64
Size_in_SqFt	int64
Price_in_Lakhs	float64
Price_per_SqFt	float64

Year_Built	int64
Furnished_Status	object (String)
Floor_No	int64
Total_Floors	int64
Age_of_Property	int64
Nearby_Schools	int64
Nearby_Hospitals	int64
Public_Transport_Accessibility	object (String)
Parking_Space	object (String)
Security	object (String)
Amenities	object (String)
Facing	object (String)
Owner_Type	object (String)
Availability_Status	object (String)

Missing Value Analysis:

An assessment using `isnull().sum()` showed that the dataset was exceptionally clean in terms of null values. All 23 columns reported exactly 0 missing values (0.0%).

Duplicate Records Check:

A check for duplicate records confirmed that there were 0 duplicate rows in the dataset.

3. Data Cleaning Steps Performed

Missing Value Handling: As zero missing values were detected across all 250,000 records, no imputation or dropping strategies were required.

Duplicate Removal: No duplicate rows were found, so the record count remained at 250,000.

Outlier Detection: We utilized the Interquartile Range (IQR) method to check for outliers on the three most critical numeric columns: Price_in_Lakhs, Size_in_SqFt, and Price_per_SqFt.

- Price_in_Lakhs (bounds: -233.9 to 743.4): 0 outliers found.
- Size_in_SqFt (bounds: -1753.5 to 7250.5): 0 outliers found.
- Price_per_SqFt (bounds: -11974.0 to 32764.2): 19,723 statistical outliers were identified (representing 7.9% of the data).

Given that luxury real estate can legitimately have very high per-square-foot prices, we elected not to drop these rows. Instead, we capped (clipped) the `Price_per_SqFt` values at the 95th percentile to handle extreme skewness while preserving valid data.

Invalid Value Handling: Basic descriptive statistics (min values) were checked. The minimum Price_in_Lakhs was 10.0, Size_in_SqFt was 500, and Floor_No was 0. No negative or zero values were found for dimensions that strictly require positive numbers.

4. Feature Validation

Price_per_SqFt Recalculation: During validation, a logical inconsistency was discovered in the original `Price_per_SqFt` column. We recalculated it using the formula: `(Price_in_Lakhs * 100000) / Size_in_SqFt`. The original sample values (e.g., 0.10, 0.08) were incorrect fractions, whereas our recalculated values correctly reflected true market rates (e.g., ₹10,332/sq.ft, ₹8,270/sq.ft). The original column was dropped in favor of a clean derivation later in the pipeline.

Logical Integrity Check: We checked if `Floor_No` was greater than `Total_Floors`. The analysis found 116,304 instances of this logical error, indicating the original synthetic data generation had a flaw. This required feature engineering adjustments in the next phase.

Dropped Columns: The columns `ID`, `Year_Built`, and `Locality` were dropped as they were deemed unnecessary for the predictive modeling phase.

5. Before vs After Comparison

Metric	Before Cleaning	After Cleaning
Total Records	250,000	250,000
Total Columns	23	20
Missing Values	0	0

6. Final Clean Dataset Summary

Final Shape: 250,000 rows × 20 columns.

Value Ranges for Key Numerical Columns:

- **Price_in_Lakhs:** Min: 10.0 | Mean: 254.58 | Max: 500.0
- **Size_in_SqFt:** Min: 500 | Mean: 2749.81 | Max: 5000
- **Age_of_Property:** Min: 2 | Mean: 18.47 | Max: 35

7. Conclusion

The data cleaning phase successfully validated the structural integrity of the 250,000 records. While the dataset was exceptionally clean regarding nulls and duplicates, crucial logical validation identified scaling issues in the price per square foot metric and logical inconsistencies in floor data. Irrelevant columns were stripped, leaving a robust, 20-feature dataset ready for Exploratory Data Analysis and Feature Engineering.